

Pizza Sales Analysis Report 2015

An End-to-End Data Analysis using SQL, Python, Excel & Tableau

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SQL | Python | Excel | Tableau

Table of Contents

1. EXECUTIVE SUMMARY	3
2. DATASET OVERVIEW	4
3. UNIVARIATE ANALYSIS	5
3.1 WHAT IS THE DISTRIBUTION OF TOTAL_PRICE?	5
3.2 WHAT IS THE DISTRIBUTION OF QUANTITY?.....	5
3.3 WHAT IS THE DISTRIBUTION OF UNIT_PRICE?.....	6
3.4 WHAT IS THE VALUE COUNT OF PIZZA_CATEGORY?	6
3.5 WHAT IS THE VALUE COUNT OF PIZZA_SIZE?.....	7
3.6 WHAT IS THE VALUE COUNT OF TIME_OF_DAY?	7
3.7 WHAT IS THE VALUE COUNT OF DAY_OF_WEEK?.....	8
3.8 WHAT IS THE DISTRIBUTION OF IS_WEEKEND?.....	8
3.9 WHAT IS THE VALUE COUNT OF IS_VEGETARIAN?	9
3.10 WHAT IS THE VALUE COUNT OF ORDER_SIZE?	9
3.11 WHAT IS THE DISTRIBUTION OF REVENUE_CATEGORY?	10
4. BIVARIATE ANALYSIS	11
4.1 WHICH PIZZA CATEGORY GENERATES THE MOST REVENUE?	11
4.2 WHICH PIZZA SIZE GENERATES THE MOST REVENUE?	11
4.3 WHAT IS THE REVENUE TREND ACROSS MONTHS?	12
4.4 WHAT IS THE REVENUE TREND ACROSS QUARTERS?	13
4.5 WHICH TIME OF DAY GENERATES THE MOST REVENUE?	13
4.6 WHAT IS THE REVENUE COMPARISON BETWEEN WEEKDAY AND WEEKEND?	14
4.7 WHAT IS THE REVENUE COMPARISON BETWEEN VEG AND NON-VEG?	14
4.8 TOP 5 AND BOTTOM 5 PIZZAS BY REVENUE?	15
4.9 TOP 5 AND BOTTOM 5 PIZZAS BY QUANTITY SOLD?	16
4.10 WHAT IS THE CORRELATION BETWEEN QUANTITY, UNIT_PRICE AND TOTAL_PRICE?	17
5. BUSINESS QUESTIONS:.....	18
5.1 WHICH DAY OF THE WEEK HAS THE HIGHEST REVENUE, QUANTITY AND ORDERS?	18
5.2 WHICH PIZZA CATEGORY HAS THE HIGHEST REVENUE, QUANTITY AND ORDERS?	18
5.3 WHICH PIZZA SIZE HAS THE HIGHEST REVENUE, QUANTITY AND ORDERS?	19
5.4 WHICH TIME OF DAY HAS THE HIGHEST REVENUE, QUANTITY AND ORDERS?.....	20
5.5 WHAT IS THE REVENUE, QUANTITY AND ORDERS COMPARISON BETWEEN WEEKDAY AND WEEKEND?	21
5.6 WHAT IS THE REVENUE, QUANTITY AND ORDERS COMPARISON BETWEEN VEG AND NON-VEG?	21
5.7 WHAT IS THE MONTHLY REVENUE, QUANTITY AND ORDERS TREND?	22
5.8 WHAT IS THE QUARTERLY REVENUE, QUANTITY AND ORDERS TREND?¶¶	23
5.9 TOP 5 AND BOTTOM 5 PIZZAS BY REVENUE, QUANTITY AND ORDERS?	24
8. RECOMMENDATIONS.....	26
9. CONCLUSION	27

1. Executive Summary

Pizza Sales Analysis 2015 examines a full year of transaction data from a pizza restaurant. The dataset contains 48,620 records across 20 columns, capturing detailed information about orders, pizza types, sizes, pricing, and customer behavior. The analysis was conducted using SQL for data extraction, Python for data cleaning and EDA, Excel and Tableau for interactive dashboards.

Key findings reveal that the business generated **\$817,860 in total revenue** from **21,350 unique orders** with **49,574 pizzas sold**. Friday is the busiest day and Lunch is the peak time of day. Classic category leads in revenue and quantity, while Large size dominates all size metrics. Thai Chicken Pizza is the top revenue generator and Brie Carre Pizza is the consistent underperformer across all metrics.

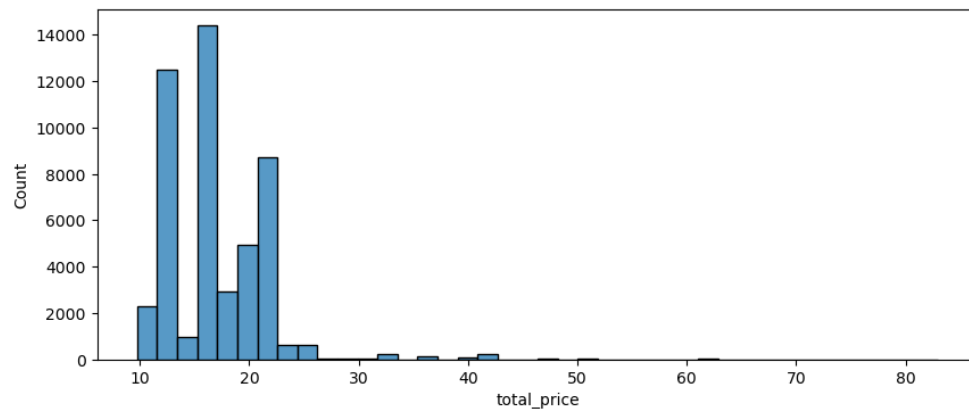
2. Dataset Overview

Column	Data Type	Description
pizza_id	int	Unique pizza record identifier
order_id	int	Unique order identifier
pizza_name_id	text	Short code/slug for the pizza
quantity	int	Number of pizzas ordered
order_date	date	Date of the order
order_time	time	Time of the order
unit_price	double	Price per pizza
total_price	double	Total price for the order
pizza_size	text	Size (S, M, L, XL, XXL)
pizza_category	text	Category (Classic, Supreme, Veggie, Chicken)
pizza_ingredients	text	List of ingredients
pizza_name	text	Full name of the pizza
day_of_week	text	Day of the week (Monday–Sunday)
is_weekend	text	Weekend or Weekday
month_name	text	Month name (January–December)
quarter	text	Quarter (Q1–Q4)
time_of_day	text	Morning / Lunch / Afternoon / Evening / Night
is_vegetarian	text	Veg or Non-Veg
order_size	text	Small / Medium / Large (based on quantity)
revenue_category	text	Budget / Standard / Premium (based on unit price)

Total Rows: 48,620 | **Total Columns:** 20 | **Time Period:** January–December 2015

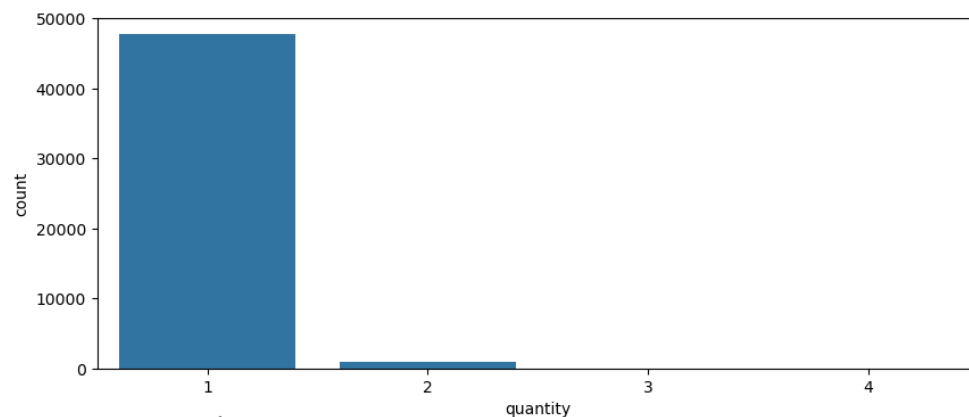
3. Univariate Analysis

3.1 What is the distribution of total_price?



- Distribution is right skewed - most of the orders between \$10 - \$25.
- Peak range \$15 - \$17 - most common price point.
- Very less orders exist in \$30+ range.
- Outliers exist in range \$50 - \$85 - multiple quantity orders.

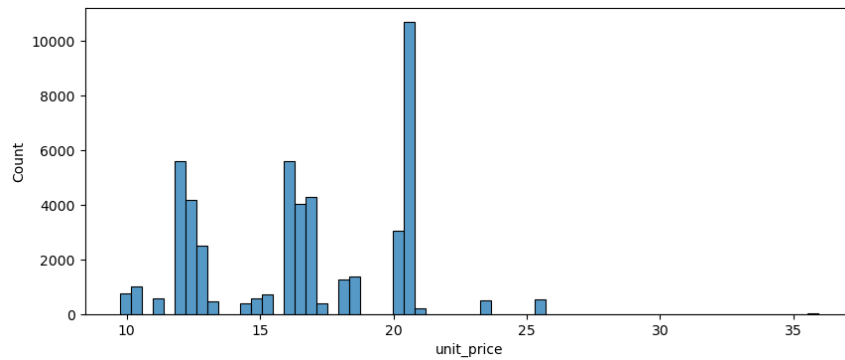
3.2 What is the distribution of quantity?



quantity	count
0	1 47693
1	2 903
2	3 21
3	4 3

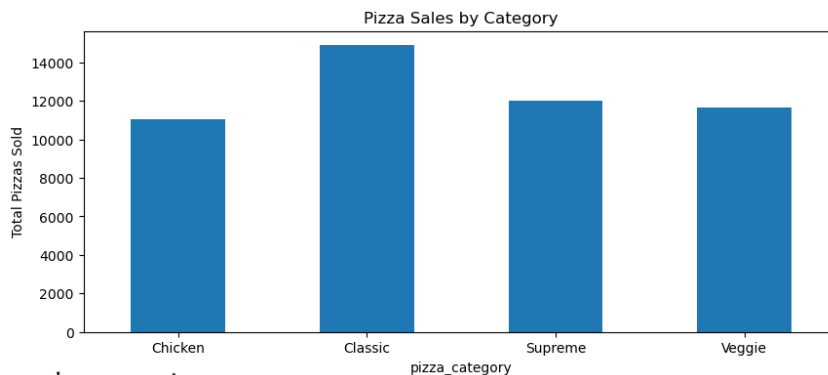
- 47,693 orders are for single pizza - most of the customers order just 1 pizza.
- 2, 3, 4 quantity very rare.
- Data heavily imbalanced - 1 quantity overwhelmingly dominant.

3.3 What is the distribution of unit_price?



- \$20.75 is the most common price - 8,685 orders
- \$12.00 second most common price - 5600 orders
- \$35.95 is least common price - only 28 orders
- Can divide into 3 cluster
 - Budget Cluster: \$9.75 - \$14.75
 - Standard Cluster: \$15.25 - \$18.50
 - Premium Cluster: \$20.25 - \$35.95

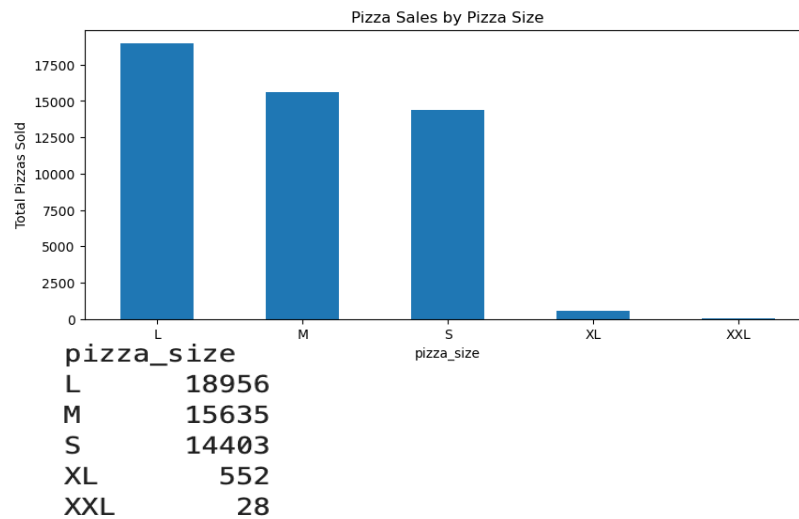
3.4 What is the value count of pizza_category?



```
pizza_category
Chicken      11050
Classic      14888
Supreme      11987
Veggie       11649
```

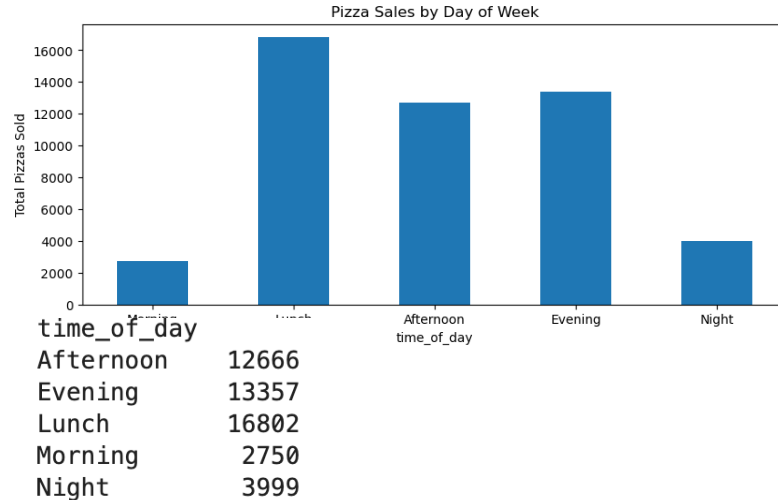
- Classic category sold highest total pizzas
- Chicken sold lowest among categories
- Category distribution relatively balanced
- Difference exists because some orders contain multiple pizzas

3.5 What is the value count of pizza_size?



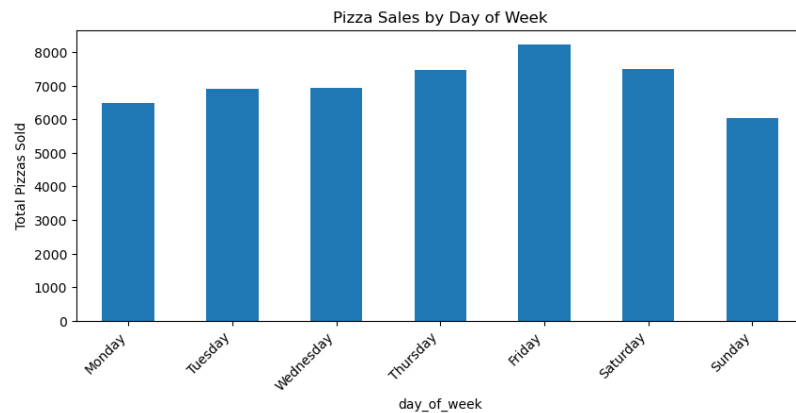
- Large(L) most popular - 18,956
- M & S near to same - balanced demand
- XL & XXL very low demand - 552 & 28
- Clear pattern: $L > M > S \gg XL \gg XXL$
- XL + XXL contribute only ~1.2% of total pizzas sold

3.6 What is the value count of time_of_day?



- Lunch is the busiest time - 16,802 pizzas sold
- Evening second - 13,357 pizzas sold
- Afternoon follows – 12,666 pizzas sold
- Night has lower demand – 3,999 pizzas sold
- Morning is the least busiest time - 2,750 (fewer people eat pizza between 9 - 11AM)
- Peak time: Lunch + Evening
- These two periods contribute the majority of sales
- Morning demand is very low (early hours, low traffic)

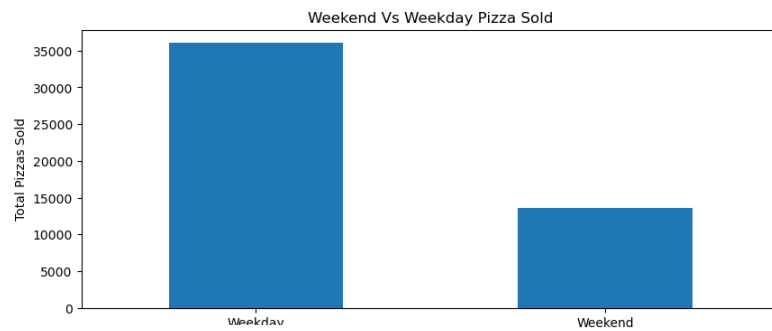
3.7 What is the value count of day_of_week?



```
day_of_week
Friday      8242
Monday      6485
Saturday    7493
Sunday      6035
Thursday    7478
Tuesday     6895
Wednesday   6946
```

- Friday records the highest pizza sales(8242 units)
- Sunday shows the lowest demand (6035)
- Weekdays remain relatively stable with no extreme variation
- Slightly higher demand observed towards the end of the week.

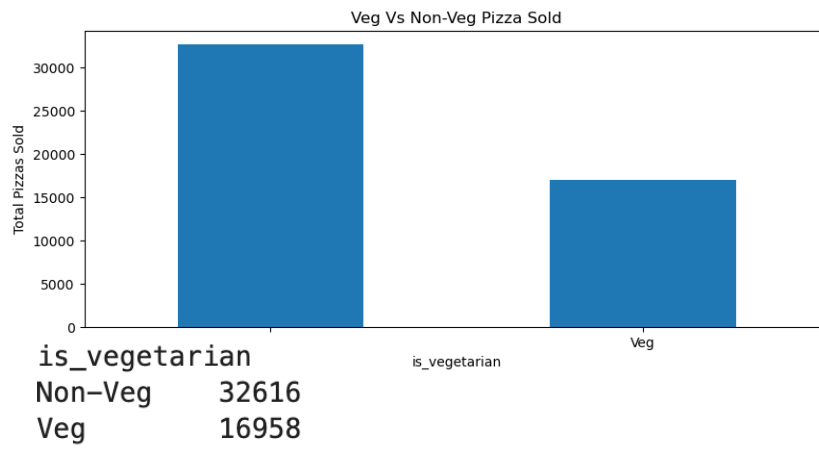
3.8 What is the distribution of is_weekend?



```
is_weekend
Weekday    36046
Weekend     13528
```

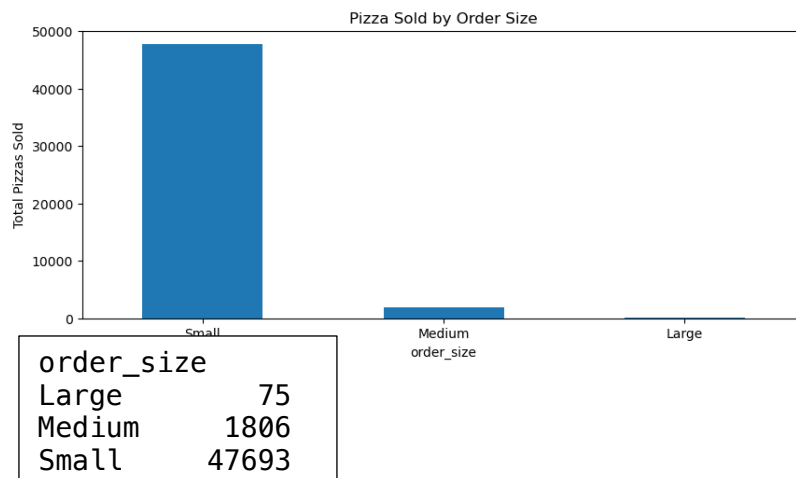
- Most of Pizzas sold in Weekday - 36,046
- In Weekend the number of pizza sold 13528 - near one third of Weekday's sold
- The reason of this Weekday = 5days and Weekend = 2days
- Per day average
 - Weekday: $36,046/5 = \sim 7209/\text{day}$
 - Weekend: $13,528/2 = \sim 6764/\text{day}$
- not a big difference on per day basis

3.9 What is the value count of `is_vegetarian`?



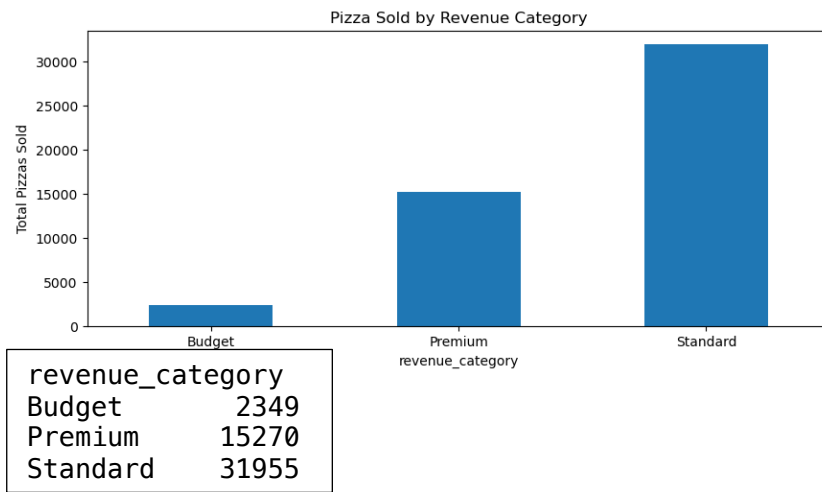
- Non-Veg most popular - 32616 pizza sold
- Veg only 16958 - nearly half of Non-Veg
- Ratio: Non-Veg 66% vs 34%
- Non-Veg dominant on the menu

3.10 What is the value count of `order_size`?



- Small (qty=1) overwhelmingly dominant — 47,693 orders
- Medium (qty=2) only 1,806
- Large (qty=3+) only 75 — rare
- consistent with the result of Q2 — most of the customer order single pizza

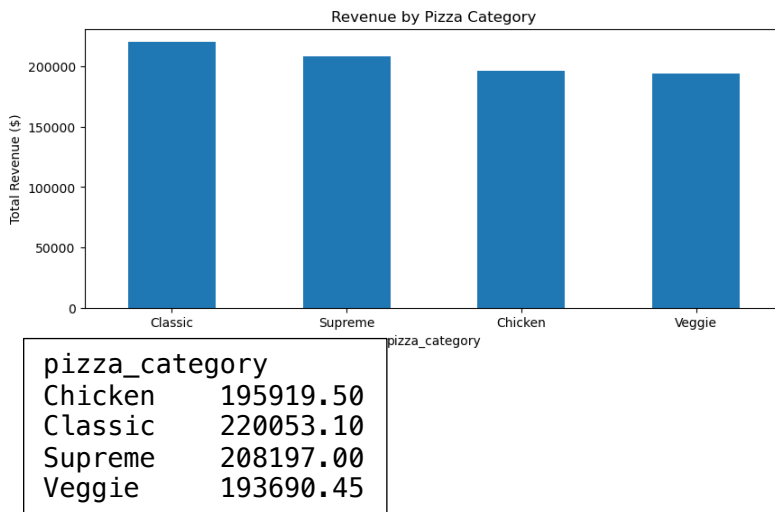
3.11 What is the distribution of revenue_category?



- Standard revenue are the most - 31955 pizzas (\$12-\$20 range)
- Premium Second - 15279 pizzas (\$20+ range)
- Budget are the least - only 2349 pizzas (under \$12 range)
- Customer most of the time spend their money for Standard & Premium range
- Budget range not the that much popular - menu could be reconsider

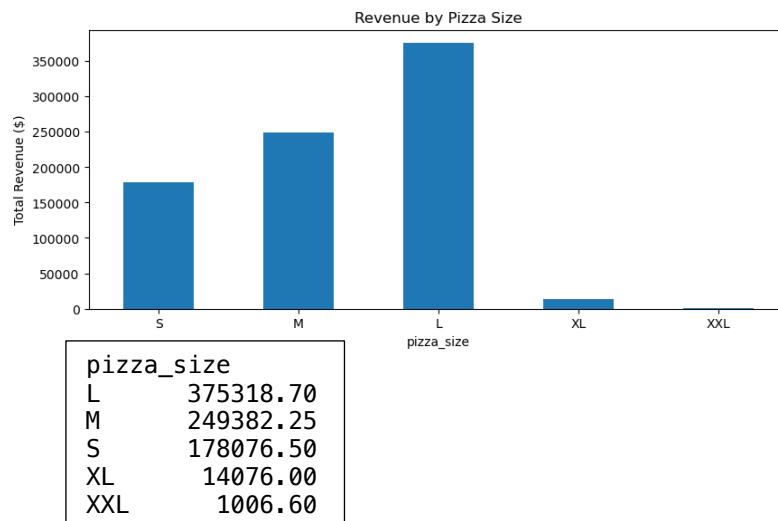
4. Bivariate Analysis

4.1 Which pizza category generates the most revenue?



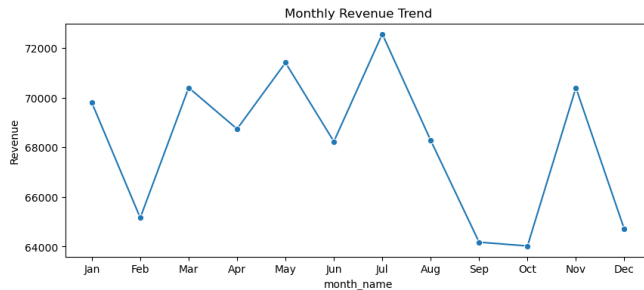
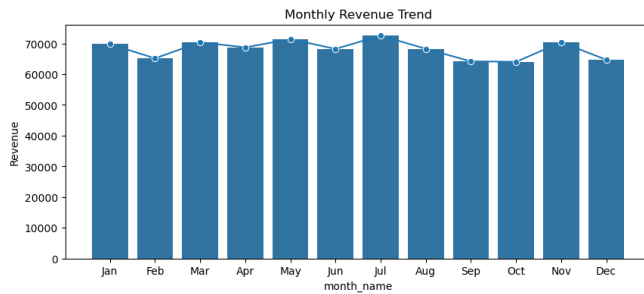
- Classic generate most of revenue — \$220,053
- Supreme second — \$208,197
- Chicken and Veggie nearly similar — ~\$194K – \$196K
- Not a big difference of revenue of all category - balanced revenue distribution
- Classic category top both quantity and revenue — consistent performer

4.2 Which pizza size generates the most revenue?



- Large(L) generate the highest revenue - \$375,318 (45.9% of total)
- -Medium is second - \$249,382
- Small/Regular(S) is third - \$178,076
- XL and XXL contribute very little - \$14,076 and \$1,006 respectively
- Clear pattern: L > M > S >> XL >> XXL
- Large size is the primary revenue driver - should be prioritized in promotions

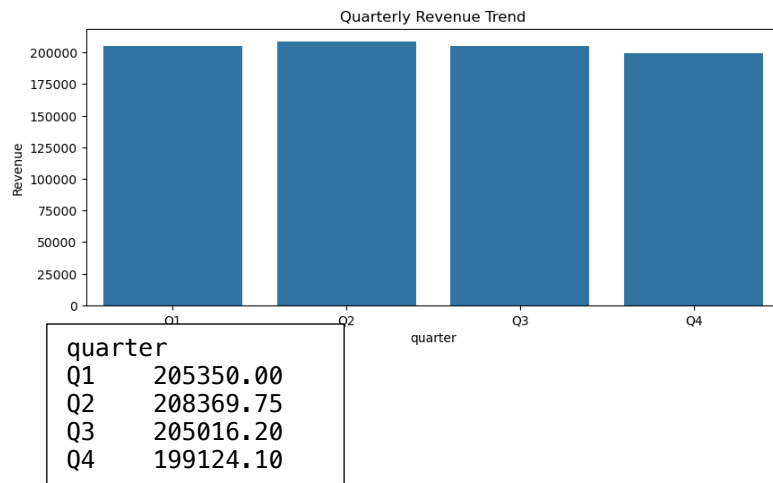
4.3 What is the revenue trend across months?



month_name	Revenue
Apr	68736.80
Aug	68278.25
Dec	64701.15
Feb	65159.60
Jan	69793.30
Jul	72557.90
Jun	68230.20
Mar	70397.10
May	71402.75
Nov	70395.35
Oct	64027.60
Sep	64180.05

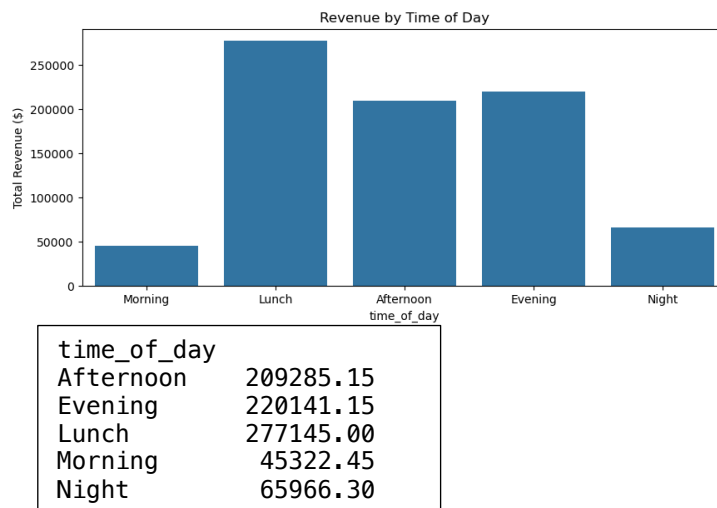
- July generated the highest revenue — \$72,557
- May and March are close second — \$71,402 and \$70,397
- October is the lowest — \$64,027
- December and September are also relatively low — ~\$64K
- Overall monthly revenue is quite consistent — ranging from \$64K to \$72K
- No dramatic peaks or drops — stable demand throughout the year
- Slight dip in Sep, Oct, Dec — could be seasonal pattern

4.4 What is the revenue trend across quarters?



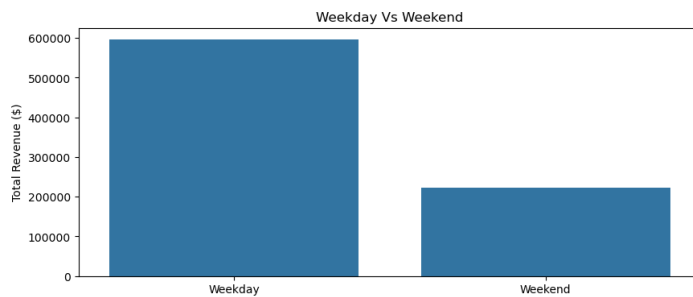
- Q2 generated the highest revenue — \$208,369
- Q1 and Q3 are very close — \$205,350 and \$205,016
- Q4 is the lowest — \$199,124
- Revenue is very consistent across all quarters — difference between highest and lowest is only ~\$9,245
- Slight decline in Q4 — Sep, Oct, Dec were the weakest months
- Business has stable year-round demand with no major seasonal fluctuations

4.5 Which time of day generates the most revenue?



- Lunch generates the highest revenue — \$277,145 (33.8% of total)
- Evening is second — \$220,141
- Afternoon is third — \$209,285
- Morning is the lowest — \$45,322 (only 5.5% of total)
- Night is also low — \$65,966
- Lunch + Evening combined = ~\$497K — over 60% of total revenue
- Business should focus staffing and promotions during Lunch and Evening hours

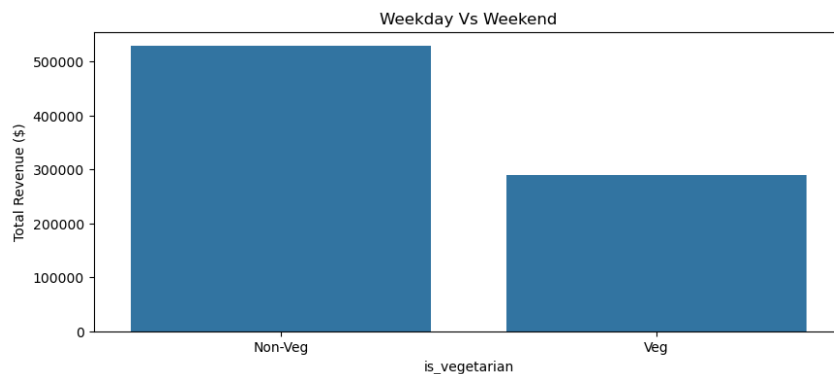
4.6 What is the revenue comparison between Weekday and Weekend?



```
is_weekend
Weekday    595474.15
Weekend    222385.90
```

- Weekday generates significantly more revenue — \$595,474 (72.8% of total)
- Weekend contributes — \$222,386 (27.2% of total)
- But per day average:
 - Weekday: $\$595,474 \div 5 = \$119,095/\text{day}$
 - Weekend: $\$222,386 \div 2 = \$111,193/\text{day}$
- Per day basis the difference is only ~7% — not as dramatic as total suggests
- Weekday dominates simply because there are more weekdays than weekends

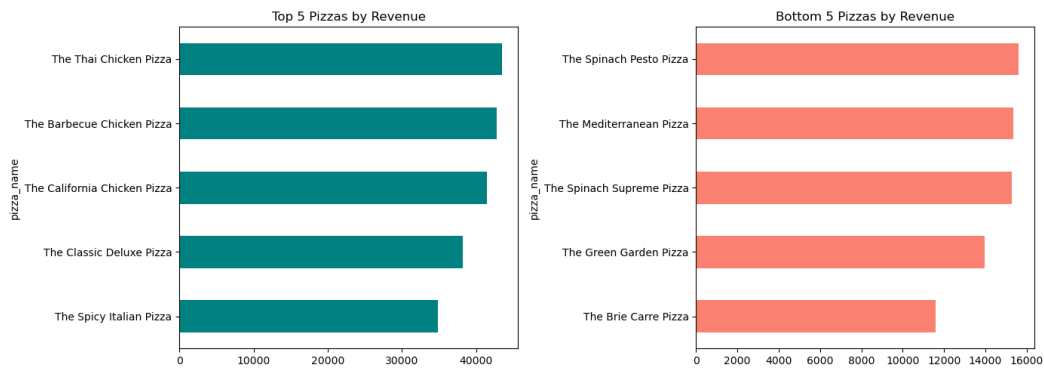
4.7 What is the revenue comparison between Veg and Non-Veg?



```
is_vegetarian
Non-Veg    528462.60
Veg        289397.45
```

- Non-Veg generates significantly more revenue — \$528,463 (64.6% of total)
- Veg contributes — \$289,397 (35.4% of total)
- Consistent with Q9 — Non-Veg is more popular in both quantity and revenue
- Non-Veg revenue is nearly 1.8x higher than Veg
- However, Veg still contributes 35% — a significant portion worth maintaining in the menu

4.8 Top 5 and Bottom 5 pizzas by revenue?



Top 5 by Revenue:

pizza_name	Revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768.00
The California Chicken Pizza	41409.50
The Classic Deluxe Pizza	38180.50
The Spicy Italian Pizza	34831.25

Bottom 5 by Revenue:

pizza_name	Revenue
The Spinach Pesto Pizza	15596.00
The Mediterranean Pizza	15360.50
The Spinach Supreme Pizza	15277.75
The Green Garden Pizza	13955.75
The Brie Carre Pizza	11588.50

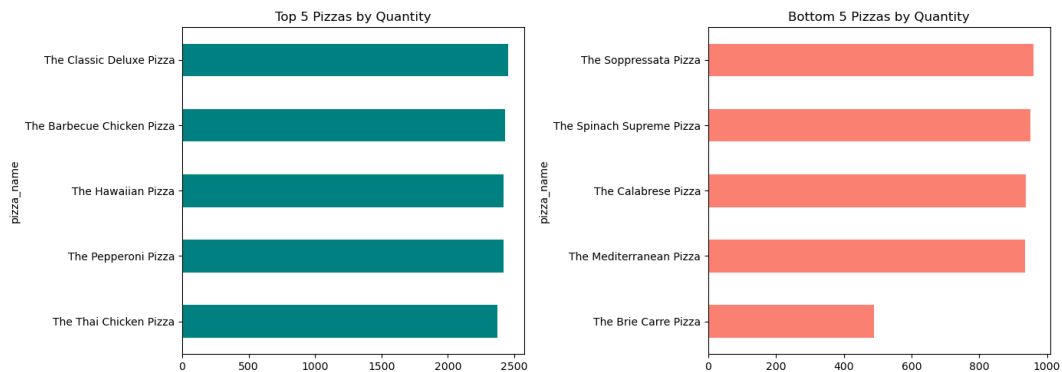
Top 5 by Revenue:

- Thai Chicken Pizza is the highest revenue generator — \$43,434
- Barbecue Chicken and California Chicken are close second and third — ~\$42K and \$41K
- 3 out of top 5 are Chicken pizzas — Chicken category dominates top revenue
- Classic Deluxe is the only Classic in top 5 — \$38,180

Bottom 5 by Revenue:

- Brie Carre Pizza is the worst performer — only \$11,588
- Green Garden Pizza is second worst — \$13,955
- All bottom 5 are Veg pizzas — consistent with overall Veg vs Non-Veg trend
- Bottom 5 revenue range: \$11K–\$15K vs Top 5: \$34K–\$43K — nearly 3x difference

4.9 Top 5 and Bottom 5 pizzas by quantity sold?



Top 5 by Quantity:

```
pizza_name
The Classic Deluxe Pizza      2453
The Barbecue Chicken Pizza    2432
The Hawaiian Pizza            2422
The Pepperoni Pizza           2418
The Thai Chicken Pizza        2371
```

Bottom 5 by Quantity:

```
pizza_name
The Soppressata Pizza         961
The Spinach Supreme Pizza     950
The Calabrese Pizza           937
The Mediterranean Pizza       934
The Brie Carre Pizza          490
```

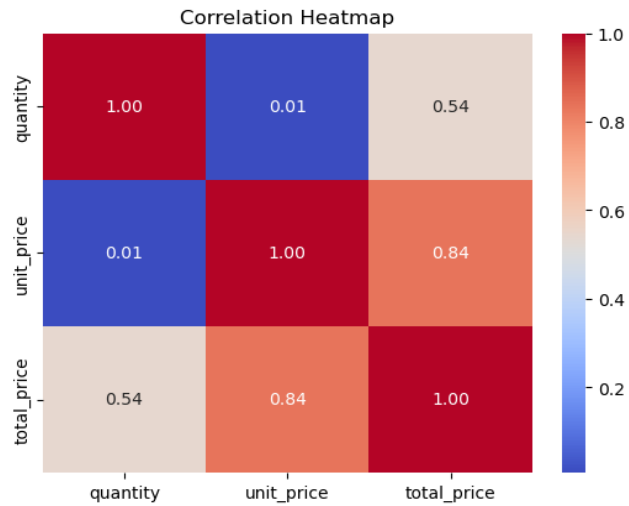
Top 5 by Quantity:

- Classic Deluxe Pizza is the most sold — 2,453 units
- Barbecue Chicken and Hawaiian are close second and third — ~2,432 and 2,422
- Top 5 are mixed categories — Classic, Chicken, and a Veggie (Hawaiian)
- Difference between top 5 is very small — highly competitive

Bottom 5 by Quantity:

- Brie Carre Pizza is the worst — only 490 units sold
- Brie Carre is significantly lower than others — nearly half of the 4th worst
- Other bottom 4 are close to each other — 934 to 961 units
- All bottom 5 are low demand pizzas — menu review recommended

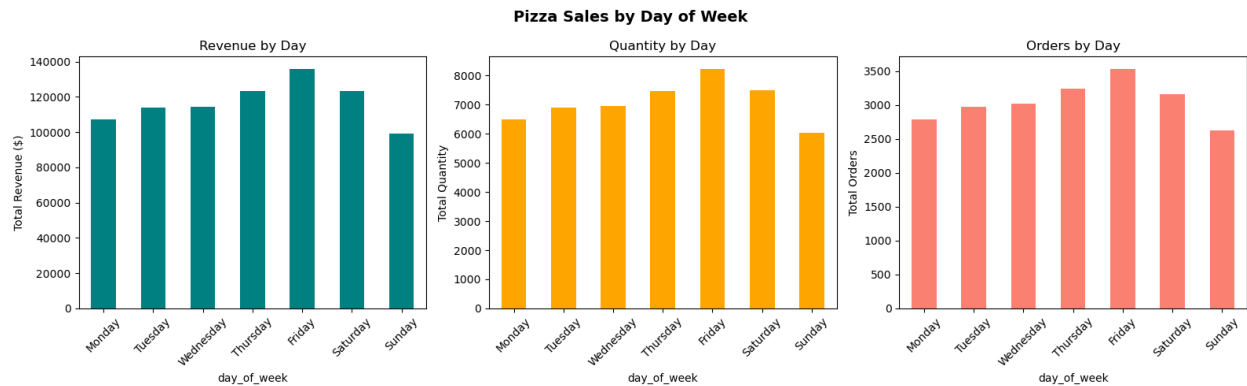
4.10 What is the correlation between quantity, unit_price and total_price?



- unit_price vs total_price: 0.84 — very strong positive correlation — higher priced pizzas generate more revenue, as expected
- quantity vs total_price: 0.54 — moderate positive correlation — more quantity means more revenue but not as strong as price
- quantity vs unit_price: 0.007 — almost zero correlation — price of pizza has no relation to how many are ordered
- Key takeaway: Price drives revenue more than quantity — premium pizzas are the bigger revenue contributors

5. Business Questions:

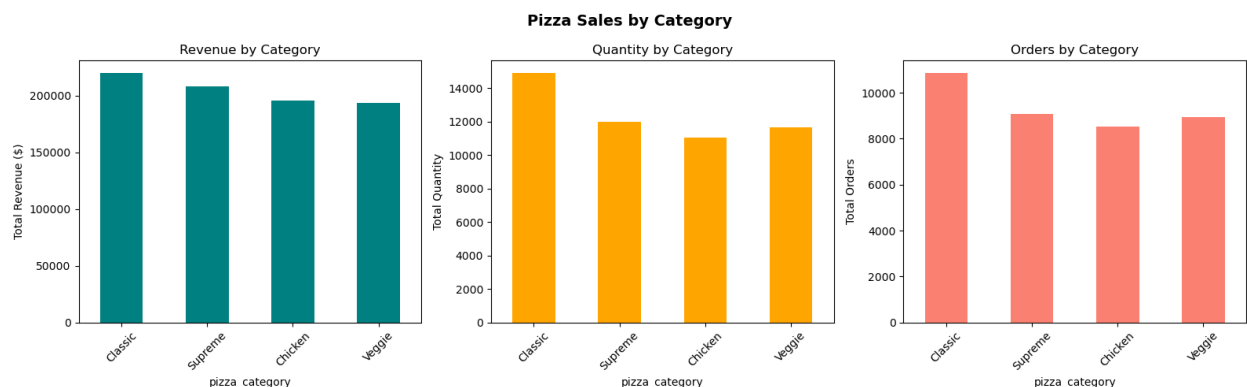
5.1 Which day of the week has the highest revenue, quantity and orders?



day_of_week	Total_Revenue	Total_Quantity	Total_Orders
Monday	107329.55	6485	2794
Tuesday	114133.80	6895	2973
Wednesday	114408.40	6946	3024
Thursday	123528.50	7478	3239
Friday	136073.90	8242	3538
Saturday	123182.40	7493	3158
Sunday	99203.50	6035	2624

- Friday leads in all 3 metrics — Revenue: \$136,073 | Quantity: 8,242 | Orders: 3,538
- Thursday and Saturday are close second — both ~\$123K revenue
- Sunday is the weakest day across all metrics — Revenue: \$99,203 | Orders: 2,624
- Monday is second weakest — Revenue: \$107,329
- Thu–Sat = peak days consistently across revenue, quantity and orders
- Sun–Mon = slowest days — promotions or discounts could help boost sales

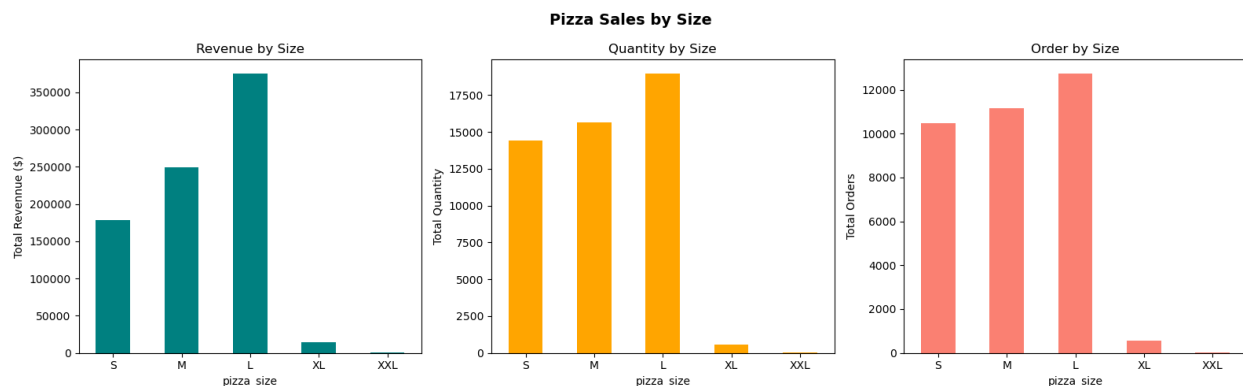
5.2 Which pizza category has the highest revenue, quantity and orders?



pizza_category	Total_Revenue	Total_Quantity	Total_Orders
Classic	220053.10	14888	10859
Supreme	208197.00	11987	9085
Chicken	195919.50	11050	8536
Veggie	193690.45	11649	8941

- Classic leads in all 3 metrics — Revenue: \$220,053 | Quantity: 14,888 | Orders: 10,859
- Supreme is second in revenue — \$208,197 but quantity is lower than Veggie
- Veggie sells more quantity than Chicken (11,649 vs 11,050) but generates less revenue — Veggie pizzas are cheaper
- Chicken generates more revenue than Veggie despite lower quantity — Chicken pizzas are higher priced
- All 4 categories are relatively balanced — no single category dominates overwhelmingly

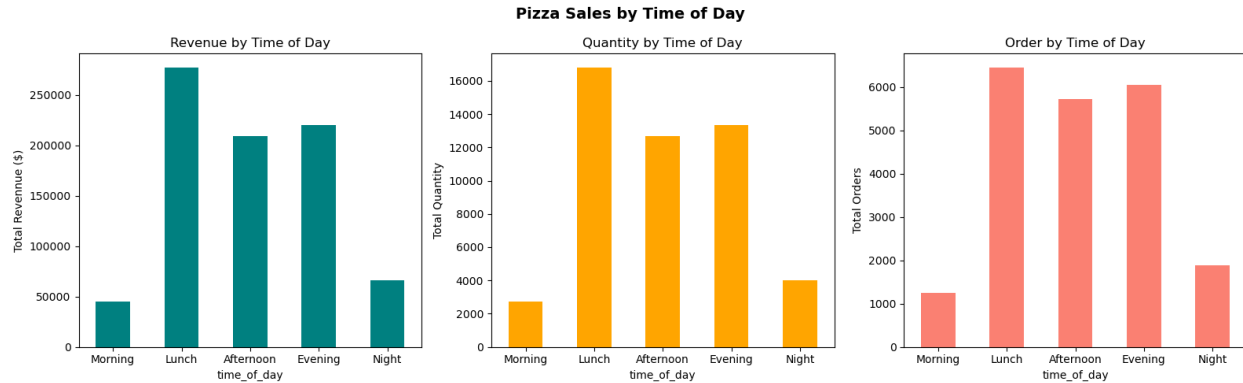
5.3 Which pizza size has the highest revenue, quantity and orders?



Total_Revenue	Total_Quantity	Total_Orders
pizza_size		
S	178076.50	14403
M	249382.25	15635
L	375318.70	18956
XL	14076.00	552
XXL	1006.60	28

- Large (L) dominates all 3 metrics — Revenue: \$375,318 | Quantity: 18,956 | Orders: 12,736
- Medium (M) is consistently second across all metrics
- Small (S) is third — despite being cheaper, still generates \$178K revenue
- XL and XXL are negligible — combined revenue only \$15,082 (1.8% of total)
- Clear pattern: L > M > S >> XL >> XXL across all metrics
- Business should focus on L and M sizes for maximum revenue impact

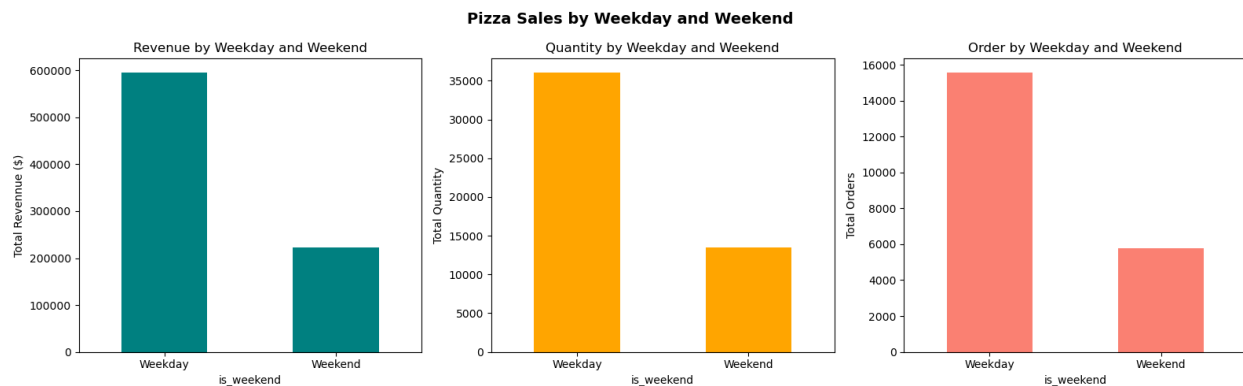
5.4 Which time of day has the highest revenue, quantity and orders?



Total_Revenue	Total_Quantity	Total_Orders	
time_of_day			
Morning	45322.45	2750	1240
Lunch	277145.00	16802	6447
Afternoon	209285.15	12666	5724
Evening	220141.15	13357	6050
Night	65966.30	3999	1889

- Lunch dominates all 3 metrics — Revenue: \$277,145 | Quantity: 16,802 | Orders: 6,447
- Evening is second — Revenue: \$220,141 | Orders: 6,050
- Afternoon is third — Revenue: \$209,285
- Morning is the weakest — Revenue: \$45,322 | Orders: only 1,240
- Lunch + Evening combined = ~\$497K — over 60% of total revenue
- Morning and Night together = only \$111K — just 13.5% of total
- Staff and promotions should be heavily focused on Lunch and Evening hours

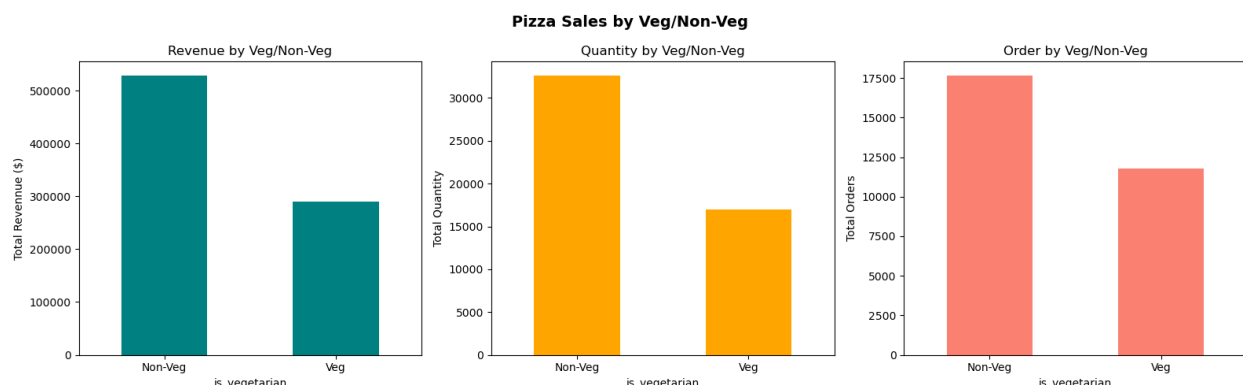
5.5 What is the revenue, quantity and orders comparison between Weekday and Weekend?



Total_Revenue	Total_Quantity	Total_Orders	
is_weekend			
Weekday	595474.15	36046	15568
Weekend	222385.90	13528	5782

- Weekday dominates all 3 metrics — Revenue: \$595,474 | Quantity: 36,046 | Orders: 15,568
- Weekend contributes — Revenue: \$222,386 | Quantity: 13,528 | Orders: 5,782
- But per day average:
 - Weekday: $\$595,474 \div 5 = \$119,095/\text{day}$
 - Weekend: $\$222,386 \div 2 = \$111,193/\text{day}$
- Per day difference is only ~7% — demand is fairly consistent daily
- Weekday dominates in totals simply because there are more weekdays

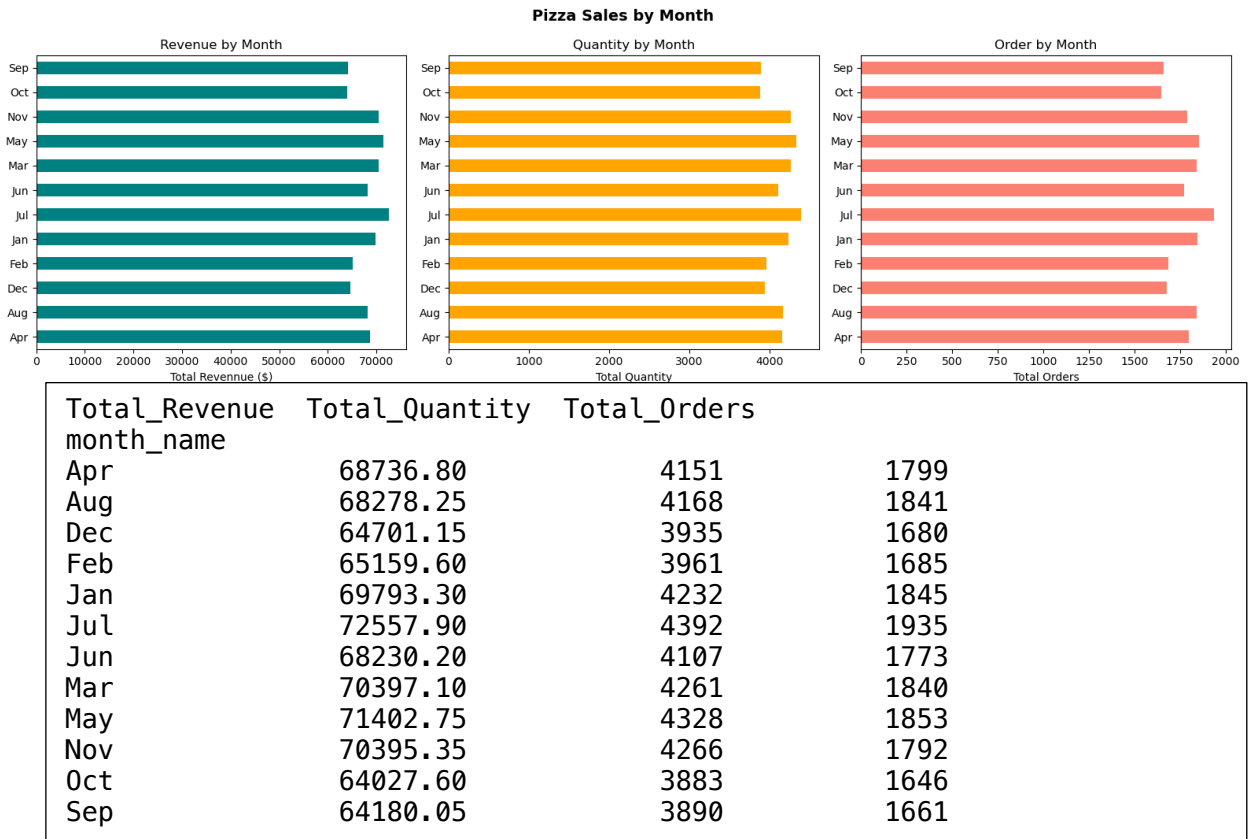
5.6 What is the revenue, quantity and orders comparison between Veg and Non-Veg?



Total_Revenue	Total_Quantity	Total_Orders	
is_vegetarian			
Non-Veg	528462.60	32616	17655
Veg	289397.45	16958	11766

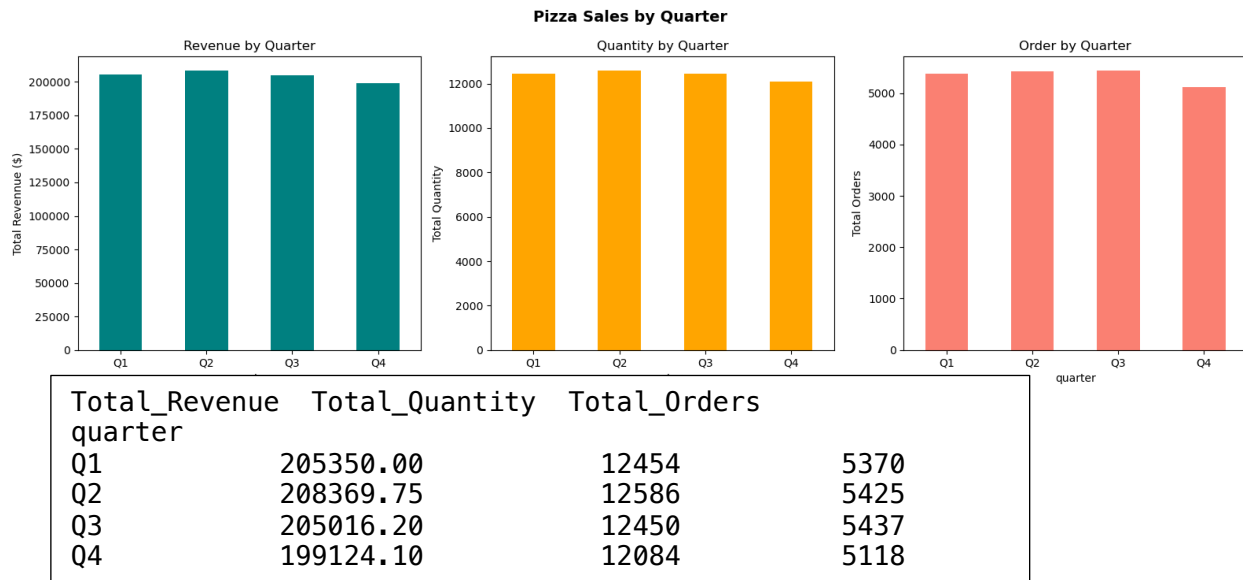
- Non-Veg dominates all 3 metrics — Revenue: \$528,463 | Quantity: 32,616 | Orders: 17,655
- Veg contributes — Revenue: \$289,397 | Quantity: 16,958 | Orders: 11,766
- Non-Veg revenue is 1.8x higher than Veg
- Non-Veg accounts for 64.6% of total revenue
- Veg still contributes 35.4% — a significant portion worth maintaining
- Menu should prioritize Non-Veg but not neglect Veg options

5.7 What is the monthly revenue, quantity and orders trend?



- July is the best month across all metrics — Revenue: \$72,557 | Quantity: 4,392 | Orders: 1,935
- May and March are close second — ~\$70K–\$71K revenue
- October is the weakest month — Revenue: \$64,027 | Orders: 1,646
- December and February are also low — ~\$64K–\$65K
- Overall monthly trend is very consistent — range is only \$64K–\$72K
- Summer months (May–July) perform slightly better
- Q4 months (Oct, Nov, Dec) show slight decline — possible seasonal dip

5.8 What is the quarterly revenue, quantity and orders trend?



- Q2 is the best quarter across all metrics — Revenue: \$208,369 | Quantity: 12,586 | Orders: 5,425
- Q1 and Q3 are very close — Revenue ~\$205K, Quantity ~12,450
- Q4 is the weakest quarter — Revenue: \$199,124 | Quantity: 12,084 | Orders: 5,118
- Difference between best (Q2) and worst (Q4) is only ~\$9,245 — very stable
- Orders in Q3 slightly higher than Q1 despite lower revenue — Q3 has slightly cheaper orders
- Business maintains consistent performance year-round with a slight Q4 dip

5.9 Top 5 and Bottom 5 pizzas by revenue, quantity and orders?

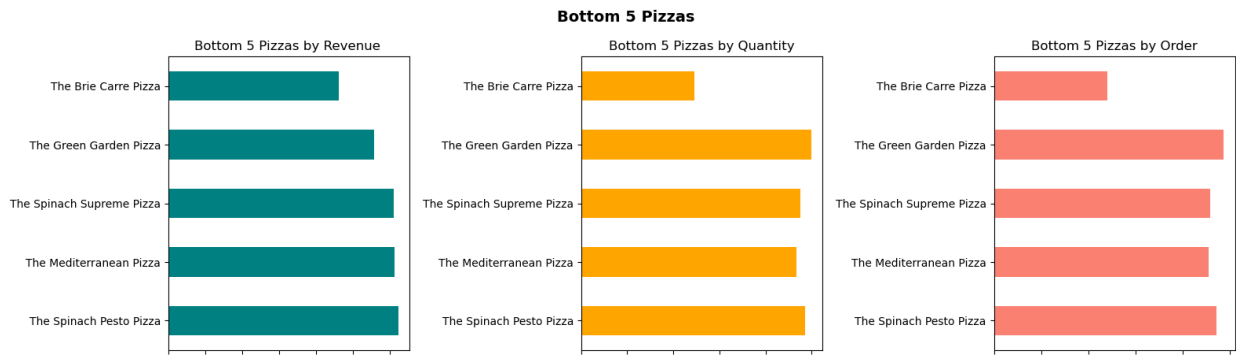
Top 5 Pizzas:



Top 5 Pizzas		Total_Revenue	Total_Quantity	Total_Orders
pizza_name				
The Thai Chicken Pizza		43434.25	2371	2225
The Barbecue Chicken Pizza		42768.00	2432	2273
The California Chicken Pizza		41409.50	2370	2197
The Classic Deluxe Pizza		38180.50	2453	2329
The Spicy Italian Pizza		34831.25	1924	1822

- Thai Chicken Pizza leads in revenue — \$43,434 despite not being #1 in quantity
- Classic Deluxe has the highest quantity (2,453) and orders (2,329) but only 4th in revenue — it's a lower priced pizza
- Top 3 are all Chicken pizzas — Chicken category dominates revenue
- Spicy Italian is 5th in revenue but only 1,924 in quantity — it's a higher priced pizza

Bottom 5 Pizzas:



Bottom 5 Pizzas		Total_Revenue	Total_Quantity	Total_Ord
ers				
pizza_name				
The Spinach Pesto Pizza		15596.00	970	945
The Mediterranean Pizza		15360.50	934	912
The Spinach Supreme Pizza		15277.75	950	918
The Green Garden Pizza		13955.75	997	976
The Brie Carre Pizza		11588.50	490	480

- Brie Carre Pizza is the worst across all 3 metrics — Revenue: \$11,588 | Quantity: 490 | Orders: 480
- Brie Carre is significantly worse than others — nearly half the quantity of other bottom pizzas
- Green Garden has highest quantity (997) among bottom 5 but lowest revenue after Brie Carre — very cheap pizza
- All bottom 5 are Veg pizzas — consistent with overall Veg underperformance

8. Recommendations

1. Focus on Peak Days and Hours Friday, Thursday, and Saturday generate the highest revenue. The business should increase staffing, run special promotions, and ensure full inventory on these days. Lunch (12–2 PM) and Evening (6–8 PM) are the busiest hours — additional staff and faster service during these windows will maximize revenue.

2. Prioritize Large Size Pizzas Large size dominates all metrics — 45.89% of total revenue. Marketing campaigns and combo deals should be built around Large size to further boost its share.

3. Revamp or Remove Underperforming Pizzas Brie Carre Pizza consistently ranks last in revenue (\$11,588), quantity (490), and orders (480). The business should consider removing it from the menu or redesigning it with more popular ingredients. Green Garden and Mediterranean pizzas also show weak performance and need review.

4. Leverage Chicken Category 3 out of top 5 revenue-generating pizzas are Chicken pizzas. The business should expand the Chicken category with new variants and promote them heavily as they command higher prices and strong demand.

5. Address Q4 Decline October, November, and December show the weakest performance across all metrics. Seasonal promotions, holiday combos, and discount offers during Q4 could help recover lost revenue.

6. Boost Morning and Night Sales Morning (9–11 AM) and Night (9–11 PM) together contribute only 13.5% of total revenue. Introducing breakfast pizza deals or late-night offers could unlock untapped revenue potential.

7. Promote Veg Pizzas Strategically Veg pizzas contribute 35.4% of revenue — a significant share. Targeting vegetarian customers through dedicated promotions or veg combo deals could increase this segment's contribution.

9. Conclusion

This end-to-end analysis of Pizza Sales 2015 successfully transformed raw transaction data into actionable business insights using SQL, Python, Excel, and Tableau.

The business demonstrates **stable year-round demand** with total revenue of \$817,860 from 21,350 unique orders. Classic category and Large size are the consistent top performers across all metrics. Chicken pizzas dominate the revenue leaderboard despite lower quantity — indicating strong pricing power. Friday Lunch hours represent the single most valuable business window.

Key areas of concern include the consistent underperformance of Brie Carre Pizza, a slight Q4 revenue dip, and the very low contribution of Morning and Night hours. Addressing these gaps through targeted promotions and menu optimization presents a clear opportunity for revenue growth.

Overall, the pizza business is healthy and well-balanced, with no single point of failure — but significant upside remains in optimizing peak performance windows and addressing weak spots in the menu and calendar.